

NILS JÖRGENSEN

Robotics Researcher & Systems Engineer | Smart Factory | Edge AI & Fleet Automation

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EXPERIENCE

Ph. D. Researcher | Networked Industrial Robotics

KTH Royal Institute of Technology

📅 Autumn 2020 – Spring 2026 📍 Stockholm, Sweden

Hands-on research at the intersection of smart factory automation, multi-robot coordination, and industrial network infrastructure. From the factory-specific communication middlewares, through edge computing architectures and digital twins, up to fleet-level task planning and mission coordination for mobile robotic systems. Broad understanding of where AI fits and where it might not, with a clear-eyed view of the emerging transition from rule-based to context-based automation in production environments. Equally at home with system architecture as with implementation: how to represent factory assets and their constraints, what belongs at the edge, and how sensing, communication, reasoning, and execution interact as a coupled system rather than independent layers.

Teacher, Student Supervisor, and Research Engineer

KTH | Department of Engineering Design

📅 Autumn 2019 – Spring 2026 📍 Stockholm, Sweden

Designed and delivered courses on embedded control systems, robotics applications, modeling and control of dynamical systems, and mechatronics in general. Supervised degree projects and industrial proof-of-concept projects through the Mechatronics capstone course, spanning the full range from embedded control and sensor integration to industrial automation prototypes.

EDUCATION

Ph. D. in Mechatronics and Embedded Control Systems

KTH Royal Institute of Technology

📅 Autumn 2020 – Spring 2026 📍 Stockholm, Sweden

- Trustworthy Cyber-Physical Systems
- Fault-Tolerant Design
- Emerging Factory Automation
- Networked Control Systems
- Reinforcement Learning
- Optimization
- Scientific Writing

M. Sc. Eng. in Engineering Design – Mechatronics Track

KTH Royal Institute of Technology / RWTH Aachen University

📅 Autumn 2017 – Spring 2019 📍 Stockholm / Aachen, Germany

- Machine Design
- Manufacturing Technology
- Electronics
- Systems Engineering
- Programming
- Embedded Systems
- Numerical Analysis
- Nonlinear Control
- Internetworking

LANGUAGES

Swedish

English

German



TECHNICAL SKILLS

-  **Smart Factory architecture**
End-to-end automation design for Industry 4.0: AGV/AMR, private 5G, digital twins, edge/cloud & IIoT.
-  **Multi-robot coordination**
Task allocation, QoS-aware replanning, and joint scheduling of tasks and shared resources in structured factory environments.
-  **AI, learning and optimization**
Pragmatic application of machine learning and edge AI where it demonstrably improves outcomes.
-  **MATLAB, Python, C++ and ROS2**
Prototyping and deployment across the full stack. Simulation and hardware-in-the-loop validation.

PUBLICATIONS

-  **Joint 5G-aware task planner for industrial multi-robot use-case**
10.1109/ETFA54631.2023.10275420
-  **Survey of communication-aware planning algorithms for factory 5G**
10.1109/ETFA52439.2022.9921449
-  **Robotic measurement campaign & Gaussian process model validation**
10.48550/arXiv.2603.08865
-  **Trustworthy edge computing for CPS and Industry 4.0: a review**
10.1145/3539662

REFERENCES

Available upon request.